ECE 382M-7 | WEEKLY TEAM EXECUTION PLAN

Hardware-Accelerated Monte Carlo Option Pricing
October 18 - December 9, 2026

Project: 1/2/4-lane European call-option accelerator. **Repository:** github.com/chuanqichen/Hardware-Accelerated-Monte-Carlo-Simulation

Professor feedback: Define numerical accuracy criteria and provide convincing evidence that the throughput target is achievable.

Key milestones

Date	Milestone	Required evidence
Nov 10	Intermediate checkpoint	Working one-lane RTL; bit-accurate verification; preliminary synthesis and timing
Dec 1 / 3	Presentations	Architecture, accuracy, throughput, scaling, and PPA results
Dec 9	Final report	1/2/4-lane study; reproducible sources; representative physical-design evidence

Team ownership

Owner	Responsibility	Primary deliverables
Tyler	Top-level, synthesis, physical design	Controller, mc_top, SDC, synthesis/P&R, PPA
Chuanqi	Gaussian LUT and fixed point	gauss_lut, round_sat, formats, accuracy, LUT PPA
Eric	Pricing and accumulation	MAC, exp LUT, payoff, accumulator, reference
Shentong	Random number generation	xoshiro128++, independent seeds, stream tests

Toolchain

Tool	Use	Owners
Git/GitHub	Branches, pull requests, reviews	All
Python + NumPy	Golden model, vectors, accuracy, Black-Scholes	All
VCS + DVE/Verdi	RTL simulation and waveform debugging	All
Verilator	RTL lint	All
Design Compiler + SDC	Cell synthesis, timing, area	Tyler + owners
Course-supported P&R / STA / power tools	Placement, routing, timing, power	Tyler
Matplotlib, Pandoc/LaTeX	Plots and documentation	All

Confirm the physical-design, timing, and power tool licenses on the course lab machines. The repository explicitly identifies VCS and Design Compiler; additional tools are proposed, not yet verified.

Phase I: Intermediate checkpoint (Oct 18 - Nov 10)

Week 1 | Oct 18-24 | Freeze interfaces and verify standalone blocks

Owner	Tasks
Chuanqi	Run Python accuracy sweep; VCS Gaussian LATENCY=1/2; implement round_sat.sv and mc_pkg.sv; test signed ties and saturation; prepare 8/10/12-bit LUT variants; agree on Q-formats.
Shentong	Implement xoshiro128++ RNG, seed load/reset/run/u_valid; compare software sequences; reject all-zero state; prove one word per cycle.
Eric	Implement B*z, A+Bz, bounded exponential approximation, payoff and accumulation; determine latency and exp range; generate corner vectors.
Tyler	Freeze mc_top configuration interface; controller skeleton; create SDC and Design Compiler scripts; verify technology libraries and trial synthesis.

Exit gate: All owners approve lane interface; standalone tests pass; representative block synthesizes; latency/valid contracts documented.

Week 2 | Oct 25-31 | Integrate one lane

Owner	Tasks
Chuanqi	Wire u to gauss_lut to z; verify valid alignment; contribute Gaussian function to bit-accurate Python lane model.
Shentong	Connect RNG and Gaussian; verify deterministic stream and uninterrupted input valid.
Eric	Finish pricing pipeline and accumulator; compare raw intermediate values against Python.
Tyler	Integrate controller and lane; implement configuration, ISSUE/DRAIN/DONE, cycle_count and status.

Test cases: reset and seed; 100/1,000/10,000 trials; exact payoff_sum/trial_count comparison; repeated identical seed; continuous p_valid after fill; saturation and invalid-configuration checks.

Exit gate: One lane has zero unexplained RTL-vs-Python mismatches.

Week 3 | Nov 1-7 | Accuracy and synthesis feasibility

Owner	Tasks
Chuanqi	Synthesize Gaussian ADDR_BITS=8/10/12; record area and critical delay; generate LUT accuracy-vs-area chart.
Shentong	Synthesize RNG; report one-word/cycle throughput, area, and timing.
Eric	Synthesize price datapath; analyze multiplier/exp critical path; pipeline if required without changing II.
Tyler	Synthesize full one-lane design; report cell area, Fmax estimate, critical path, and timing constraints.

Accuracy: Separate LUT distribution bias, fixed-point/exponential numerical bias, and finite-sample Monte Carlo noise. Proposed target: systematic relative price bias at most 1% across an agreed representative benchmark grid, with 95% confidence intervals for Monte Carlo estimates. Use absolute error for near-zero reference prices.

Throughput: Verify initiation interval (II)=1 by simulation, and check clock feasibility using synthesized critical-path timing. A waveform alone does not establish frequency.

Week 4 | Nov 8-10 | Freeze checkpoint evidence

• One-lane block diagram and configuration interface

• Bit-exact RTL-vs-Python test logs and waveforms

• Gaussian/RNG accuracy and pricing benchmark table

• Initial area, timing, and critical-path reports

• Sustained valid-output evidence and remaining timing risks

• Checkpoint slides: Tyler architecture/PPA; Shentong RNG; Chuanqi Gaussian; Eric pricing

Checkpoint definition of done: Correct one-lane implementation with quantified numerical error and credible initial synthesis timing. Routed layout is not required by the team proposal at this checkpoint.

Phase II: Scaling and physical evaluation (Nov 11 - Dec 9)

Week 5 | Nov 11-17 | Expand to 2 and 4 lanes

Owner	Tasks
Tyler	Parameterize LANES=1/2/4; implement reduction tree, final-beat lane mask, and controller drain.
Chuanqi	Verify simultaneous Gaussian outputs, valid alignment, and per-lane numerical correctness.
Eric	Verify multi-lane accumulation, reduction latency, overflow and status flags.
Shentong	Generate independent lane seeds and validate streams and reproducibility.

Exit gate: All lane counts match their own Python reference; test N_TRIALS=101, 1001, 10003; trial_count exactly equals requested count.

Week 6 | Nov 18-24 | Complete numerical/PPA tradeoff study

Configuration	LANES	ADDR_BITS	Goal
A	1	10	Baseline
B	2	10	Parallel scaling
C	4	10	Parallel scaling
D	1	8	Smaller LUT
E	1	12	Higher accuracy

Measure synthesized cell area, timing/Fmax, switching-based power if available, trials/sec, energy/trial, option-price bias, and Monte Carlo confidence intervals. Use identical libraries, process corners, timing constraints and workload definitions. Repeat 8/12-bit LUTs at 4 lanes only if time permits.

Chuanqi deliverable: Bias/KS error vs. actual synthesized Gaussian area. ROM logical storage (2/8/32 Kbit for 8/10/12 address bits with 16-bit outputs) is not equivalent to standard-cell area.

Exit gate: Freeze core numerical and synthesis results by Nov 24.

Week 7 | Nov 25-30 | Representative physical implementation

• Tyler: import synthesized netlist and standard-cell libraries; apply SDC; floorplan and power plan.

• Tyler: placement optimization, clock-tree synthesis, routing, post-route timing and physical checks.

• Tyler: extract parasitics; record routed area, timing, power and layout screenshots.

• Chuanqi: finalize Gaussian accuracy, format documentation, and area tradeoff.

• Eric: finalize pricing and integrated benchmark evidence.

• Shentong: finalize RNG stream quality and seed methodology.

Prioritize one representative design for place-and-route, as stated in the proposal. If routed timing does not close, disclose the violation and achievable clock period; do not substitute synthesis timing for post-route timing.

Week 8 | Dec 1-6 | Presentations and report assembly

Presentations: Dec 1 and Dec 3. Tyler presents architecture/scaling/PPA; Shentong RNG; Chuanqi Gaussian and precision tradeoff; Eric pricing and end-to-end correctness.

After presenting, consolidate plots, verify every result against raw tool reports, review the report, and avoid major unverified RTL changes.

Week 9 | Dec 7-9 | Reproducibility and submission

• Final specification, design, user, and test documentation

• Synthesizable RTL, generated ROM, and bit-accurate Python reference

• Self-checking testbenches, logs, and reproducible commands

• Synthesis/SDC/physical scripts and raw reports

• Accuracy, throughput, area, power, and energy comparisons

• Representative layout and post-route evidence

• Final report and presentation artifacts

Recommended weekly coordination

When	Purpose
Monday (20-30 min)	Assign tasks, confirm interfaces and dependencies, resolve blockers.
Thursday (30-45 min)	Live demo of passing tests, review timing, decide PR merges.
Sunday (asynchronous)	Update GitHub issues with evidence and next-week risks.

Critical risks and fallback priorities

Risk	Response
Exp LUT complexity	Implement correct bounded approximation before optimizing.
Mismatch between modules	Diff raw u, z, bz, x, S_T, p against Python trace.
Timing fails	Pipeline the critical stage while preserving II=1.
4-lane area too large	Keep 1/2-lane results and report the limitation.
Physical closure fails	Report measured violation and achievable lower clock.
Price bias too high	Restrict supported inputs or increase LUT resolution.

Chuanqi: immediate daily checklist

• Mon Oct 19: Python unit tests, sweep, VCS Gaussian LATENCY=1 and 2.

• Tue Oct 20: round_sat.sv and directed signed rounding/saturation tests.

• Wed Oct 21: mc_pkg.sv and Q-format review with Eric.

• Thu Oct 22: baseline Gaussian synthesis with Tyler.

• Fri Oct 23: compare LUT 8/10/12-bit accuracy and synthesis.

• Weekend Oct 24-25: integration-ready PR and evidence summary.

Guiding priority: Before Nov 10: functional RTL -> bit-exact verification -> II=1 evidence -> synthesis timing. After Nov 10: 1/2/4-lane scaling -> representative physical implementation -> PPA/accuracy tradeoffs -> final documentation.

Source note: Milestone dates, ownership and scope follow the revised GitHub proposal. Weekly assignments, suggested acceptance thresholds and tool selections are planning recommendations. Verify any additional course submission/layout requirements against the official project brief.